

ALEX CHEN

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EDUCATION

University of Southern California

Los Angeles, CA

Bachelor of Science in Computer Engineering and Computer Science | GPA: 3.93

May 2027

Relevant Coursework: System on Chip, MOS VLSI Circuit Design, Computer Organization, Parallel Computing, Internetworking

TECHNICAL SKILLS

Design Tools: Cadence OrCAD Capture CIS, Virtuoso, Allegro, Synopsys Laker, Altium | Languages: Verilog, C++, Python, MATLAB

Core Competencies: RTL design, signal integrity, custom layout, FPGA, timing closure

EXPERIENCE

HQ Telecom

Shanghai, China

Hardware Engineer Intern

May 2026 – August 2026

- Designed schematics for HP's OmniBook Ultra laptop in Cadence OrCAD Capture CIS, achieving 150 mW Modern Standby power
- Resolved 50+ EE design issues by reworking components down to 0201 and retesting with oscilloscopes/Ellisys analyzers
- Delivered 10-layer PCB with USB4/Thunderbolt, LPDDR5X, and Nova Lake CPU at production-level SNR and crosstalk margins

Originsilicon Integrated Circuit Company

Los Angeles, CA

R&D Intern

June 2024 – September 2024

- Implemented 8+ custom analog layout blocks in Synopsys Laker for mobile LCD display driver ICs through tape-out
- Designed level shifters, output buffers, and switch arrays using common-centroid matching, guard rings, and dummy insertion
- Cut channel block height by up to 50% through floorplan restructuring, achieving clean DRC/LVS checks on basic unit blocks

PROJECTS

PawStation Smart Pet Feeder

Los Angeles, CA

Embedded Systems & Electrical Design

February 2026 – July 2026

- Built smart pet feeder using RPi, TMC2209-driven NEMA 17 stepper, and HX711 load-cell feedback for closed-loop dispensing
- Designed full power and interface stack from wall input to motor drivers, I²C/SPI peripherals, and DS3231 RTC scheduling
- Integrated REST API and MJPEG camera stream for mobile status tracking, feed scheduling, and history data analysis

In-Car Heads-Up Display System

Los Angeles, CA

Embedded Systems & Automotive Data Integration

November 2025 – February 2026

- Built Raspberry Pi-based in-car display overlaying Google Maps with real-time vehicle telemetry from OBD-II ports
- Implemented Python pipeline for ECU data parsing, synchronization, and low-latency updates on HDMI touchscreen
- Optimized the interface for glance-based driving with responsive rendering and clear real-time data display

Multiply-and-Accumulate (MAC) Accelerator

Los Angeles, CA

Circuit Design & Arithmetic Unit Development

April 2025 – May 2025

- Engineered 16-bit MAC datapath in 45nm CMOS, meeting timing closure at 0.33 GHz with radix-4 Booth encoding
- Designed custom standard-cell library and verified functionality with Cadence Virtuoso, Spectre simulation, and DRC/LVS signoff
- Optimized 6-to-2 compressor tree for delay, power, and area, reducing critical-path delay 20%

Blackjack Game on Artix-7 FPGA

Los Angeles, CA

Digital Design & RTL Implementation

January 2025 – April 2025

- Designed Verilog RTL game logic on Artix-7 FPGA with a multi-state FSM controlling deal, hit/stand, and scoring flow
- Engineered hardware pseudo-random card generation using a maximal-length LFSR seeded from a free-running counter sampled on user input, using human reaction timing as an entropy source for non-repeating deals across power cycles

RESEARCH EXPERIENCE

Shofflab

Cleveland, OH

Research Assistant

February 2023 – September 2024

- Co-authored peer-reviewed Biomaterials publication on reducing intracortical microelectrode inflammation, cited 16+ times
- Measured implanted microelectrode arrays via electrochemical impedance spectroscopy (EIS), analyzed degradation in MATLAB
- Automated data-processing pipeline in Python/Pandas, cutting analysis time by 2 hours per experiment across 80GB of datasets

LEADERSHIP & ACTIVITIES

Member, Association for Computing Machinery at USC

March 2025 – Present

Member, Taiwanese American Organization at USC

September 2024 – Present